Source-Frozen Rank-Midpoint Contrasts for the Literal V59 Scalar

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Abstract

We freeze a nontrivial two-block direction from the ordered physical interval of the literal V59 scalar before inspecting any coefficient, margin, or sign. The first $\lfloor N/2 \rfloor$ coordinates and the remaining coordinates define a normalized Haar contrast z. Its rank-one projector admits the exact rational representation $\rho^2 h_i h_j$, even when ρ is irrational. We derive exact partial-sum formulas for both coarse and midpoint longitudinal terms. Their difference is $\overline{\langle z, w\rangle}\langle z, A_x\beta\rangle$, with the first-slot conjugation explicit, and the transverse covariance changes by its negative; the remainder is the sum of the two within-child covariances. For integral x, the rank split is exactly the threshold $\lfloor 3x/4 \rfloor$. Substitution of the literal TPC-247 kernel retains its prime weight, two unit masks, deleted diagonal, orientation, physical kernel, centered residue bracket, and β. The valid transfer is $\langle z, A_x\beta\rangle = \langle A_x^* z, \beta\rangle$; self-adjointness is not used. A canonical exact certificate, 59 rejected mutations, and 192 exact-rational stress families reproduce these identities. Constant and signed controls show that the compiler alone proves no sign, nonzero value, or arithmetic scale.

1 Frozen source choice and scope

Keep the finite literal TPC-247 source object

$$g = A_x\beta, \qquad C_x = \langle w, g\rangle, \tag{1}$$

on $I_x = (x/2, x] \cap \mathbb{Z}$, with the inner product conjugate-linear in its first slot [2]. TPC-252 shows that a declared binary refinement adds one orthogonal contrast direction and transfers its covariance from the transverse to the longitudinal term [1]. Unrestricted partition optimization ends at the degenerate singleton partition. The narrow next question is therefore different: can the physical coordinates themselves freeze one nontrivial split before any coefficient is observed?

Let

$$I_x = \{n_1 < \cdots < n_N\}, \quad N \geq 2, \qquad \ell = \lfloor N/2 \rfloor, \quad r = N - \ell, \tag{2}$$

and set $L = \{n_1, \ldots, n_\ell\}$ and $R = \{n_{\ell+1}, \ldots, n_N\}$. This choice uses only x and the ordered coordinate set. It is declared before $\beta, w, A_x\beta$, any margin, or any sign. It is a source-frozen modeling choice, not a unique V59-canonical partition and not V59's smooth bounded-overlap mesoscopic partition.

The exact release status is

`PROVED_STRUCTURAL_L1_SOURCE_FROZEN_RANK_MIDPOINT_CONTRAST_COMPILER`.

2 Rank-midpoint geometry

Put

$$h = \frac{\mathbf{1}_L}{\ell} - \frac{\mathbf{1}_R}{r}, \qquad \rho^2 = \frac{\ell r}{N}, \qquad z = \rho h. \tag{3}$$

Proposition 1 (Normalization and projector)**.** *The vector z is the unique unit child-flat zero-sum contrast whose sign is positive on L. If $u_J = |J|^{-1/2}\mathbf{1}_J$, then*

$$M_{\mathrm{c}} = u_{I_x} \otimes u_{I_x}, \qquad M_{\mathrm{m}} = u_L \otimes u_L + u_R \otimes u_R = M_{\mathrm{c}} + z \otimes z. \tag{4}$$

Every entry of the added projector is the rational number $(z \otimes z)_{ij} = \rho^2 h_i h_j$.

Proof. The cardinalities give

$$\sum_{n\in I_x} h(n) = 1 - 1 = 0, \qquad \|z\|^2 = \frac{\ell r}{N}\left(\frac{1}{\ell} + \frac{1}{r}\right) = 1.$$

If a child-flat zero-sum unit vector equals $a > 0$ on L and $b < 0$ on R, then $b = -\ell a/r$ and unit norm forces $a = \sqrt{r/(\ell N)} = \rho/\ell$; hence $b = -\rho/r$. Thus the sign convention fixes z uniquely. The orthonormal pair u_{I_x}, z spans the same child-flat plane as u_L, u_R, proving (4). Finally $z_i\overline{z_j} = \rho^2 h_i h_j$ because h is real. □

The rational projector representation is the executable normalization: no floating approximation to ρ is needed when N is odd.

Proposition 2 (Integer crosswalk)**.** *For integral $x = k \geq 3$, the last coordinate of L is $\lfloor 3k/4 \rfloor$. Equivalently,*

$$L = (k/2, \lfloor 3k/4 \rfloor] \cap \mathbb{Z}, \qquad R = (\lfloor 3k/4 \rfloor, k] \cap \mathbb{Z}.$$

Proof. Here $I_k = \{\lfloor k/2 \rfloor + 1, \dots, k\}$ and $N = k - \lfloor k/2 \rfloor$. Write $k = 4m + s$. The four exact cases are

s	$\lfloor k/2 \rfloor$	N	ℓ	left endpoint
0	$2m$	$2m$	m	$3m$
1	$2m$	$2m+1$	m	$3m$
2	$2m+1$	$2m+1$	m	$3m+1$
3	$2m+1$	$2m+2$	$m+1$	$3m+2$

The last column is $\lfloor 3k/4 \rfloor$ in each row. □

Proposition 2 is only an integer crosswalk. Definition (2), rather than a threshold formula, remains primary for arbitrary real x.

3 Partial-sum covariance compiler

For $J \subset I_x$, write $S_f(J) = \sum_{n\in J} f(n)$ and $\mu_f(J) = S_f(J)/|J|$. Put $W_J = S_w(J)$ and $G_J = S_g(J)$.

Theorem 3 (Exact midpoint compiler)**.** *The contrast moment, two longitudinal terms, covariance transfer, and transverse terms satisfy*

$$\langle z, f\rangle = \rho\left(\frac{S_f(L)}{\ell} - \frac{S_f(R)}{r}\right), \tag{5}$$

$$C_{\text{long}}(\mathrm{m}) = \frac{\overline{W_L}G_L}{\ell} + \frac{\overline{W_R}G_R}{r}, \tag{6}$$

$$C_{\text{long}}(\mathrm{c}) = \frac{\overline{W_L + W_R}(G_L + G_R)}{N}, \tag{7}$$

$$\begin{aligned} C_{\text{long}}(\mathrm{m}) - C_{\text{long}}(\mathrm{c}) &= \overline{\langle z, w\rangle}\langle z, g\rangle \\ &= \frac{\ell r}{N}\overline{\left(\frac{W_L}{\ell} - \frac{W_R}{r}\right)}\left(\frac{G_L}{\ell} - \frac{G_R}{r}\right), \end{aligned} \tag{8}$$

$$Q_{\text{trans}}(\mathrm{m}) - Q_{\text{trans}}(\mathrm{c}) = -\overline{\langle z, w\rangle}\langle z, g\rangle. \tag{9}$$

Moreover,

$$Q_{\text{trans}}(\mathrm{m}) = \sum_{J\in\{L,R\}}\sum_{n\in J}\overline{w(n) - \mu_w(J)}\,[g(n) - \mu_g(J)], \tag{10}$$

and

$$C_x = C_{\text{long}}(\mathrm{m}) + Q_{\text{trans}}(\mathrm{m}), \qquad Q_{\text{trans}}(\mathrm{c}) = \overline{\langle z, w\rangle}\langle z, g\rangle + Q_{\text{trans}}(\mathrm{m}). \tag{11}$$

Proof. Equation (5) follows by summing the two constant values of z. Orthogonal block averaging gives (6) and (7). Proposition 1 and orthogonality of the coarse range with z give

$$\langle M_{\mathrm{m}}w, M_{\mathrm{m}}g\rangle = \langle M_{\mathrm{c}}w, M_{\mathrm{c}}g\rangle + \overline{\langle z, w\rangle}\langle z, g\rangle,$$

where the conjugation is forced by the first-slot convention. Substituting (5) gives (8). The fixed scalar C_x has the orthogonal decompositions $C_{\mathrm{long}}(\mathrm{c}) + Q_{\mathrm{trans}}(\mathrm{c})$ and $C_{\mathrm{long}}(\mathrm{m}) + Q_{\mathrm{trans}}(\mathrm{m})$, so subtraction gives (9). Finally $(I - M_{\mathrm{m}})w$ and $(I - M_{\mathrm{m}})g$ are the childwise centered vectors, yielding (10) and (11). $\square$

The theorem is a complex covariance identity. Neither longitudinal term nor their difference acquires a sign or monotonicity property.

4 Literal kernel expansion and safe adjoint

Retain the TPC-247 data

$$H = x^{21/32}, \quad Q = x^{1/3}, \quad \mathcal{Q}_x = \{q \text{ prime} : Q < q \le 2Q\}, \quad K_H(h) = \widehat{\psi_+}(h/H), \tag{12}$$

$$\beta(t) = \frac{\Lambda(t)}{\log t} - \sum_{\substack{d \mid t \\ d^{400} \le x^{133}}} \mu(d), \qquad w(u) = \Lambda(u+2) - b_x^{(z)}(u). \tag{13}$$

The inherited superscript in $b_x^{(z)}$ is source notation and does not identify the rank contrast. The literal operator is

$$A_x(u,t) = \mathbf{1}_{u \ne t} \sum_{q \in \mathcal{Q}_x} q \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} K_H(u-t) \left(\mathbf{1}_{u \equiv t \pmod q} - \frac{1}{q-1} \right). \tag{14}$$

Proposition 4 (Literal expansion and adjoint)**.** *The midpoint g-moment is*

$$\begin{aligned}\langle z, A_x\beta\rangle = \rho \sum_{q \in \mathcal{Q}_x} q \sum_{u,t \in I_x} \left(\frac{\mathbf{1}_L(u)}{\ell} - \frac{\mathbf{1}_R(u)}{r} \right) \mathbf{1}_{u \ne t} \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} \\ \times \beta(t) K_H(u-t) \left(\mathbf{1}_{u \equiv t \pmod q} - \frac{1}{q-1} \right).\end{aligned} \tag{15}$$

Also,

$$\langle z, A_x\beta\rangle = \langle A_x^* z, \beta\rangle, \qquad (A_x^* z)(t) = \sum_{u \in I_x} \overline{A_x(u,t)} z(u). \tag{16}$$

Proof. Insert (14) into (5) to obtain (15). No factor is rearranged or dropped. For (16), expand the finite sums:

$$\langle A_x^* z, \beta\rangle = \sum_t \overline{\sum_u \overline{A_x(u,t)} z(u)}\, \beta(t) = \sum_{u,t} \overline{z(u)} A_x(u,t) \beta(t).$$

This is $\langle z, A_x\beta\rangle$. The argument uses only the adjoint definition. $\square$

In particular, output u, input t, the outer weight q, both unit masks, the deleted diagonal, $K_H(u-t)$, the centered residue bracket, and literal $\beta(t)$ all remain visible. No symmetry of K_H, equality $A_x^* = A_x$, or self-adjointness is source-locked.

5 Sharp controls, exact audit, and limitations

Proposition 5 (Nonliteral sharp controls)**.** *If w is constant or g is constant, the transfer in* (8) *is zero. The synthetic choices $(w, g) = (z, z)$ and $(z, -z)$ give transfer $+1$ and -1, respectively.*

Proof. A constant factor has zero z-moment because $\sum z = 0$. For the signed controls, $\langle z, z\rangle = 1$, so the two products are 1 and -1. $\square$

These are exact finite Hilbert-space controls, not literal numerical V59 instances. They show that source-free geometry cannot decide sign, nonvanishing, or scale.

Executable audit. The certificate stores rational and Gaussian-rational data as canonical strings and uses only $\rho^2 h_i h_j$ for projector products. An odd-rank, nonintegral-clock fixture checks all formulas. A separately labeled nonliteral exact-kernel sample substitution checks every factor in (14); its deliberately non-self-adjoint matrix verifies that (16), rather than an $A_x^* = A_x$ shortcut, is used. The independent checker imports no producer, enforces exact integer types, and rejects 59 typed, semantic, digest, duplicate-key, nonfinite, and noncanonical mutations. The stress program verifies 192 deterministic families: 96 integer clocks, including 24 in each class modulo four, and 96 nonintegral rational clocks. Optimized and normal Python modes produce identical output. These computations reproduce finite algebra only.

Claim firewall. The rank midpoint is not V59-canonical and is not identified with a smooth V59 partition. The exact-sample kernel fixture is not an actual V59 numerical replay. We claim no A_x self-adjointness, contrast sign, nonzero value, asymptotic scale, arithmetic saving, L2 theorem, fixed-atom credit, Gate-B closure, strict 1/400, or twin-prime result. The open theorem is a source-backed joint estimate for $\langle z, w\rangle$ and $\langle z, A_x\beta\rangle$ on one common growing V59 clock.

6 Conclusion

The ordered physical interval now supplies one reproducible nontrivial contrast before the source coefficients are observed. Its projector, partial sums, covariance transfer, within-child remainder, literal kernel expansion, and adjoint orientation are exact. The compiler isolates the two literal imbalances that a future arithmetic theorem must estimate; it does not estimate them itself.

References

[1] Liang Wang. Binary refinement calculus and singleton degeneracy for declared-block v59 margins, 2026. TPC-252 project-local repository paper, August 25, 2026.

[2] Liang Wang. Literal v59 source-operator two-lane block attachment, 2026. TPC-247 project-local repository paper, August 25, 2026.